

LECTURE 1

INTRODUCTION TO

WEB MINING

LEK HSIANG HUI

OUTLINE

Introduction to Web Mining

Supervised Learning

Unsupervised Learning

Web Scraping Demo

INTRODUCTION TO WEB MINING

Introduction to
Web Mining

Why choose
this course?

Supervised
Learning

Unsupervised
Learning

Web Scraping
Demo

DATA



Companies are
generating a LOT
of data
e.g. sales,
transaction,
customer data,
etc

MORE DATA



Data is produced
not only by the
companies but
also by others
about the
companies

DATA MINING

Since everything is computerized nowadays, data is now stored in digital form (e.g. databases)

From these databases, the purpose of data mining is to look for patterns so as to discover more insights to the raw data

- Raw Data ➡ Patterns ➡ Knowledge

This knowledge will be helpful for **decision making**

HOW TO MAKE DECISION MAKING?

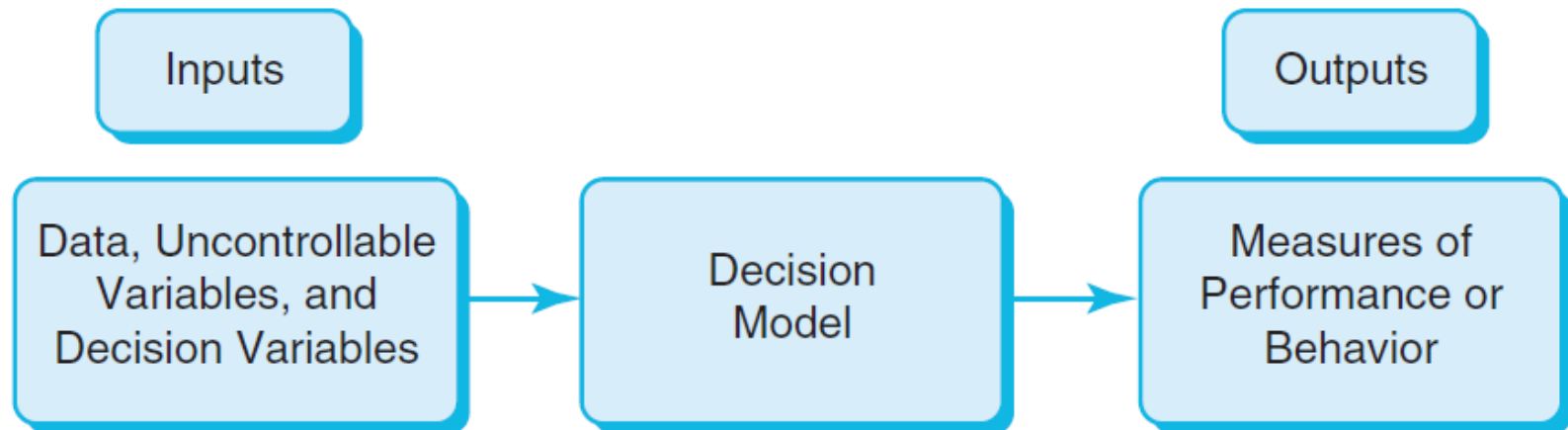
Decision Making

- Make **prediction** based on the data
 - E.g. Based on purchase behaviors (i.e. historical data) predict whether a type of product will be hot selling?
- To do this, we build **decision models** using historical data
- The models can then allow us to make prediction of future data instances

WHAT IS A DECISION MODEL?

Decision Model is a model used to understand, analyze, or facilitate decision making

- Can be in the form of a mathematical formula or software



WEB MINING

Web mining is concerned with mining data from the web which can then be transformed into knowledge

2 main aspects we will be focus on

- Mining Web Content
- Data mining techniques for handling web content (regression, classification, clustering, recommender systems)

WHY CHOOSE THIS COURSE?

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WHY THIS COURSE?

Most courses only focus on the predictive modeling aspect

- Assumes that datasets can be downloaded (e.g. from Kaggle)

However:

- The datasets we need is usually not readily available
- Having the ability to mine your own data from the web is useful skill!

>>> Becoming a Popular Bilibili Streamer: ✨ A Guide to Success ☆



Team member: Cheng Keshang, Gu Tianteng, Gong Zheng, Guo Yichen, He Haiyang
Adviser: Lek Hsiang Hui

With the popularity of live streaming platforms like Bilibili, we use web mining to analyze user engagement, channel performance, and audience behavior to provide valuable insights for content creators and enhance live streaming success.



Bilibili Analysis
--What makes vid
stand out?



Or... Do you aspire to
be a content creator?



Brooklyn Nets

Budget:

Player number:

Player Data Summary

Team Data Interface

Budget: Your budget for buying players.

Player number: The number of players you want to buy.

Search for player

Start Recommend

Name	Team	MPG	PPG	FG%	3P%	RPG	APG	SPG	BPG	TOV
Joel Embiid	PHI	34.6	33.1	0.548	0.33	10.2	4.2	1.0	1.7	3.4
Kyrie Irving	DAL	37.4	26.2	0.494	0.379	5.1	5.5	1.1	0.8	2.1
De'Aaron Fox	SAC	33.4	24.6	0.512	0.324	4.2	6.1	1.1	0.3	2.5
Julius Randle	NYK	35.5	23.2	0.459	0.343	10.0	4.1	0.6	0.3	2.8
Kevin Porter	HOU	34.3	15.7	0.442	0.366	5.3	5.7	1.4	0.3	3.2
De'Anthony Melton	PHI	27.9	10.2	0.425	0.39	4.1	2.6	1.6	0.5	1.3

Or... the best NBA manager?

Let the experts help you
decide which movie to
watch!



Douban



← → 🔍 🔍 douban.com

Film Recommendation System Based on Douban



Web Mining Group7

Ding yi, Peng Shu, Tang Jiankun, Wang Chenshen, and Ying
Jie

And being wise students... Let's
make sure we are able to detect
fake reviews on Taobao when
buying things...



SWS3023 - Opinion Spam Detection

Chen Haonan, Filbert Jonathan Hanjaya, Guo
Xinyan, Huang Yijiu, Li Shengyi

News Sentiment Analysis

上涨 rise

青睐 prefer

大涨 surge

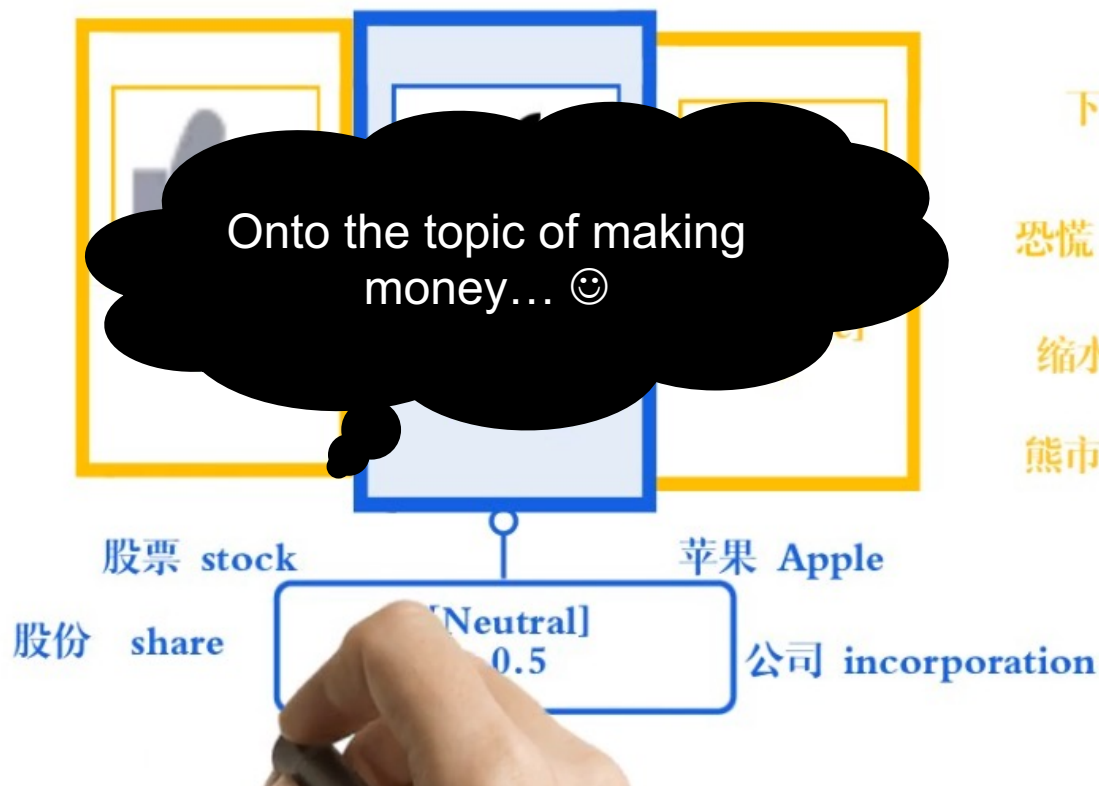
牛市 bull market

下跌 fall

恐慌 panic

缩水 shrink

熊市 bear market



Now with all those
money... it's time to buy
a house at Beijing 😊



The House Whisper



Price Prediction and Data Analysis of Second-hand Housing in **BEIJING**

MAO WEIHAO, MEI HAO, WANG BINGQIAN,
WANG ZIYUE, ZHONG ZHIYUAN



SONG RECOMMENDER SYSTEM



Group 11: Pu Shi, Chen Yuliang, Wang Rui, Ke Junlong, Liao Weikai



Music Recommendation System

Group3: Yixuan Zhao, Kexin Li, Yuheng Shi, Yingjia Li, Congkeyu Cao

Advisor: Lek Hsiang Hui



Background

Music is one of the popular categories in the digital area, we can always access the resources of many music genres and artists on internet platforms. Additionally, there are an increasing number of music applications on the market that let users rate songs and do their own searches for their personal tastes. It is crucial for recommenders to provide music lists that are extremely suitable for consumers' musical tastes while users are using music applications. Therefore, creating a music recommendation system based on user preferences is important to get a competitive edge in the music industry.



Procedures

Data Sets

Rate Your Music is a web-based platform for users to rate and score songs, covering a wide range of music albums and targeting a global user base. Music and user data is comprehensive and suitable for mining certain data to assist in modeling recommendation systems.

.Web scraping

Requests + BeautifulSoup + Proxy server

Mining the top 40 albums from the music categories, and then mining the users who have rated the albums, to get the user list of the ratings of the album data. In order to prevent the case of IP from being blocked, we at the same time used a proxy server to assist mining. Finally, we succeeded in obtaining 6.63 million ratings from 3403 different users.

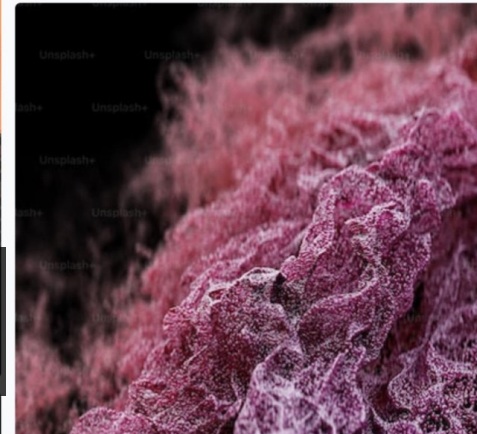
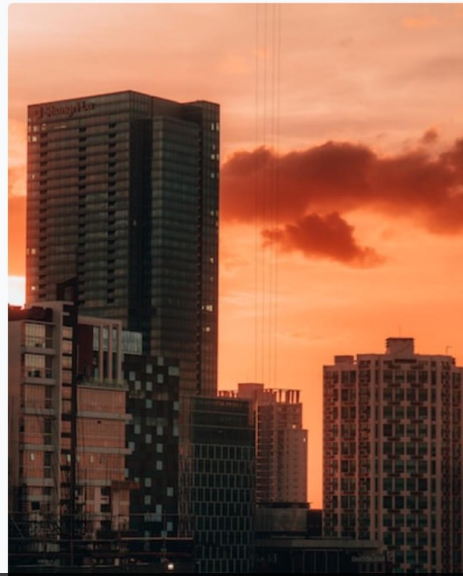
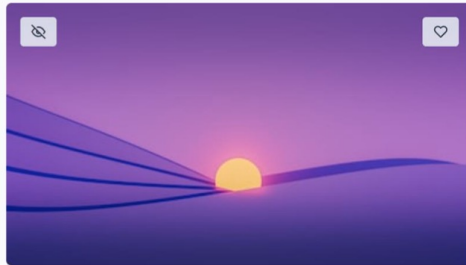
.Data cleaning

Albums rated by the same user are screened out. (There can be zero ratings for some albums, we treat them as the same.) Zero-

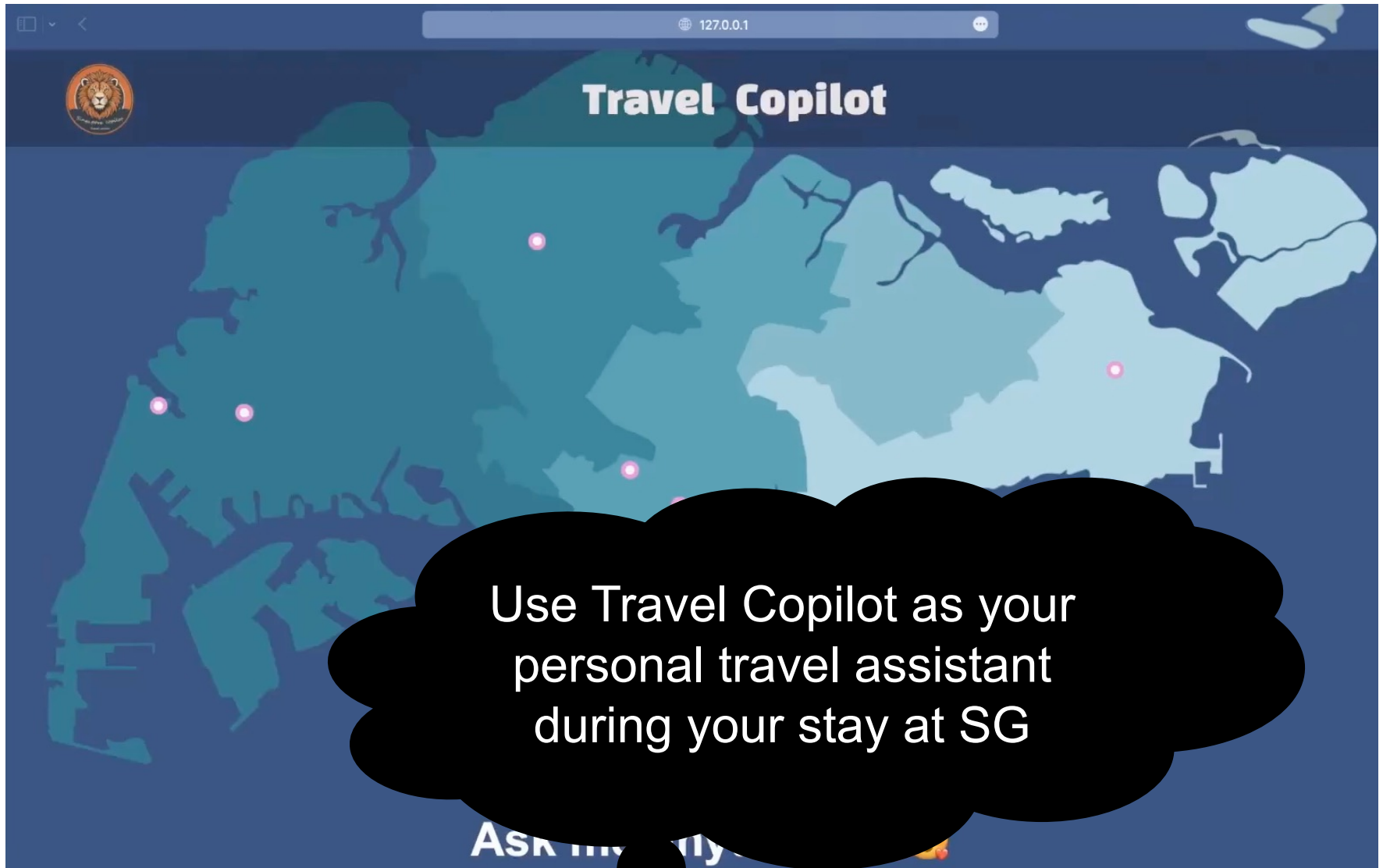


iWallpaper

Type "/" to search



Wallpaper Recommender System



CLASSIC DECISION PROBLEMS

Regression (Supervised)

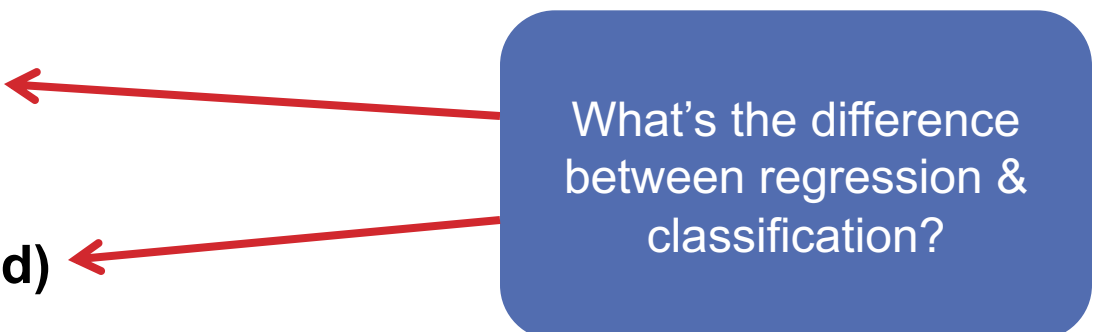
- Stock price prediction

Classification (Supervised)

- Garbage classification (recyclable, food waste, hazardous, others)

Clustering (Unsupervised)

- Group Weibo/Twitter users based on their interest



What's the difference between regression & classification?

SUPERVISED LEARNING

Introduction to
Web Mining

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Unsupervised
Learning

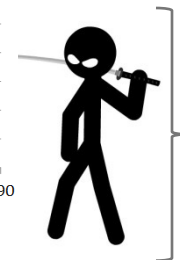
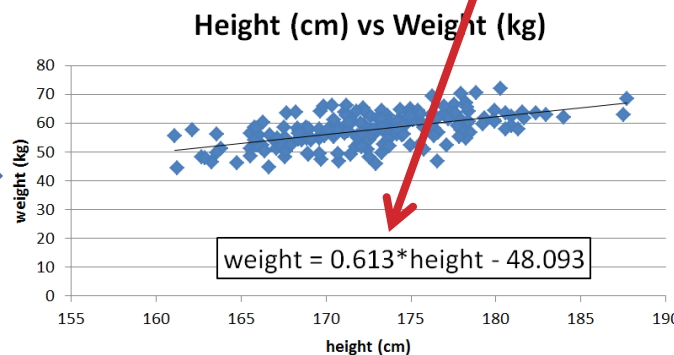
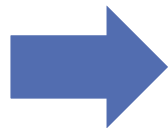
Web Scraping
Demo

REGRESSION

Regression (Supervised)

- Using existing data instances to learn a **model** for predicting subsequent instances
- Example:
 - Assume we have a list of human height and weight

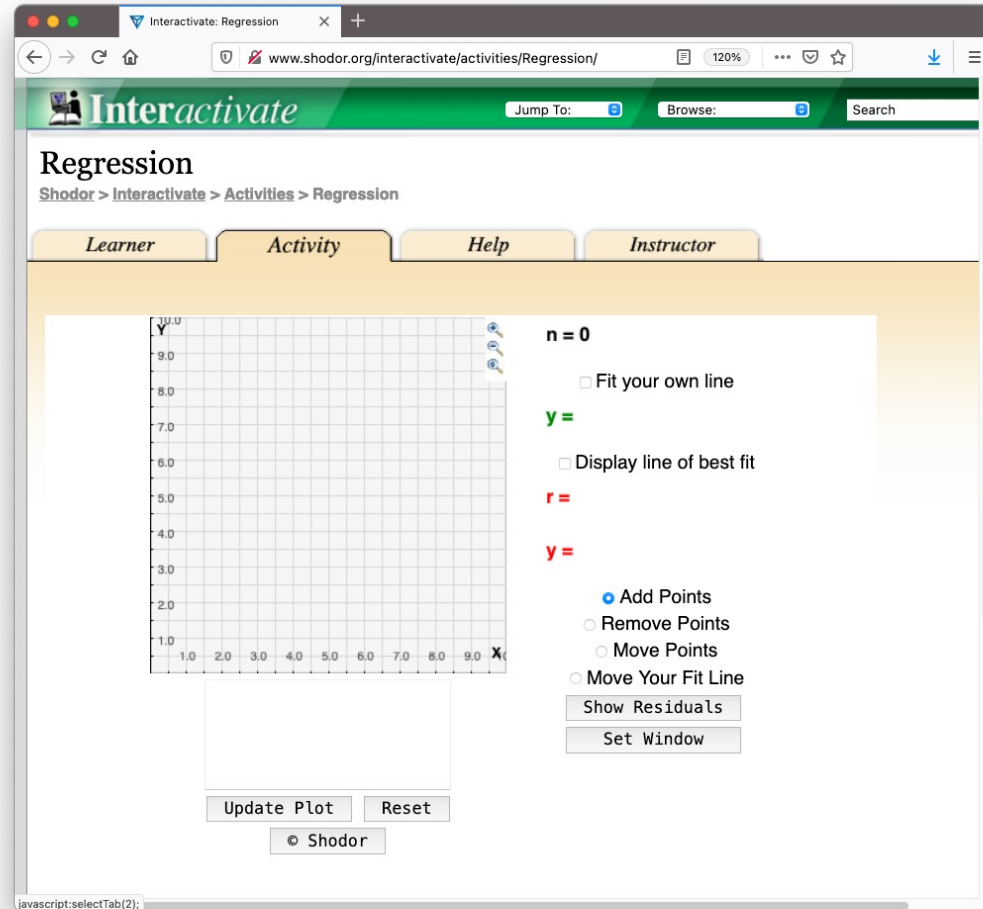
Height(cm)	Weight(kg)
167.0812	51.25136008
181.6608	61.91077208
176.276	69.41318376
173.2788	64.56428528
172.1866	65.4533256
174.498	55.9278936
177.292	64.17873208
177.8254	61.89716432
172.466	50.97013304
169.6212	54.73494664
168.8846	57.8103004
171.7548	51.77299088
173.482	56.97569112
170.4848	55.54687632
173.4312	52.65749528



height = 170cm

weight = ?

REGRESSION DEMO



<http://www.shodor.org/interactivate/activities/Regression/>

REGRESSION EXAMPLE

Dataset:

- <https://www.kaggle.com/mohansacharya/graduate-admissions>
- Prediction of Graduate Admissions from an Indian perspective
- 500 instances
- Admission_Predict_Ver1.1.csv

GRADUATE ADMISSION

	A	B	C	D	E	F	G	H	I
1	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
2	1	337	118	4	4.5	4.5	9.65	1	0.92
3	2	324	107	4	4	4.5	8.87	1	0.76
4	3	316	104	3	3	3.5	8	1	0.72
5	4	322	110	3	3.5	2.5	8.67	1	0.8
6	5	314	103	2	2	3	8.21	0	0.65
7	6	330	115	5	4.5	3	9.34	1	0.9
8	7	321	109	3	3	4	8.2	1	0.75
9	8	308	101	2	3	4	7.9	0	0.68
10	9	302	102	1	2	1.5	8	0	0.5
11	10	323	108	3	3.5	3	8.6	0	0.45
12	11	325	106	3	3.5	4	8.4	1	0.52
13	12	327	111	4	4	4.5	9	1	0.84

Identifier (unique for all instance) → Not useful for modeling

Undergraduate GPA
(out of 10)

Research Experience
(0 or 1)

Chance of Admin
(0 to 1)

GRADUATE ADMISSION

	A	B	C	D	E	F	G	H	I
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5	4	322	110	3	3.5	2.5	8.67	1	0.8
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12	11	325	106	3	3.5	4	8.4	1	0.52
13	12	327	111	4	4	4.5	9	1	0.84

GRE Scores
(out of 340)

TOEFL Score
(out of 120)

Uni. Rating
(out of 5)

Statement of Purpose
(SOP) & Letter of
Recommendation (LOR)
Recommendation Strength
(out of 5)

TYPES OF DATA

Target/Response
Dependent Variable

	A	B	C	D	E	F	G	H	I
1	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
2	1	337	118	4	4.5	4.5	9.65	1	0.92
3	2	324	107	4	4	4.5	8.87	1	0.76
4	3	316	104	3	3	3.5	8	1	0.72
5	4	322	110	3	3.5	2.5	8.67	1	0.8
6	5	314	103	2	2	3	8.21	0	0.65
7	6	330	115	5	4.5	3	9.34	1	0.9
8	7	321	109	3	3	4	8.2	1	0.75
9	8	308	101	2	3	4	7.9	0	0.68
10	9	302	102	1	2	1.5	8	0	0.5
11	10	323	108	3	3.5	3	8.6	0	0.45
12	11	325	106	3	3.5	4	8.4	1	0.52
13	12	327	111	4	4	4.5	9	1	0.84

Predictors
Independent Variables

TYPES OF DATA

Datasets with values for both predictors & response are also known as labeled data (also known as training data)

	A	B	C	D	E	F	G	H	I
1	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
2	1	337	118	4	4.5	4.5	9.65	1	0.92
3	2	324	107	4	4	4.5	8.87	1	0.76
4	3	316	104	3	3	3.5	8	1	0.72
5	4	322	110	3	3.5	2.5	8.67	1	0.8
6	5	314	103	2	2	3	8.21	0	0.65
7	6	330	115	5	4.5	3	9.34	1	0.9
8	7	321	109	3	3	4	8.2	1	0.75
9	8	308	101	2	3	4	7.9	0	0.68
10	9	302	102	1	2	1.5	8	0	0.5
11	10	323	108	3	3.5	3	8.6	0	0.45
12	11	325	106	3	3.5	4	8.4	1	0.52
13	12	327	111	4	4	4.5	9	1	0.84

TYPES OF DATA

Datasets with only values for predictors are also known as unlabeled data (also known as testing data)

	A	B	C	D	E	F	G	H	I
1	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
2	401	304	100	2	3.5	3	8.22	0	?
3	402	315	105	2	3	3	8.34	0	?
4	403	324	109	3	3.5	3	8.94	1	?
5	404	330	116	4	4	3.5	9.23	1	?
6	405	311	101	3	2	2.5	7.64	1	?
7	406	302	99	3	2.5	3	7.45	0	?
8	407	322	103	4	3	2.5	8.02	1	?
9	408	298	100	3	2.5	4	7.95	1	?
10	409	297	101	3	2	4	7.67	1	?
11	410	300	98	1	2	2.5	8.02	0	?

We want the generated model to predict the response values

SUPERVISED LEARNING

**This also illustrates the idea of
Supervised Learning**

- Teach the machine to do prediction with examples and the corresponding expected prediction

HANDS-ON: REGRESSION

HANDS-ON: REGRESSION

Dataset:

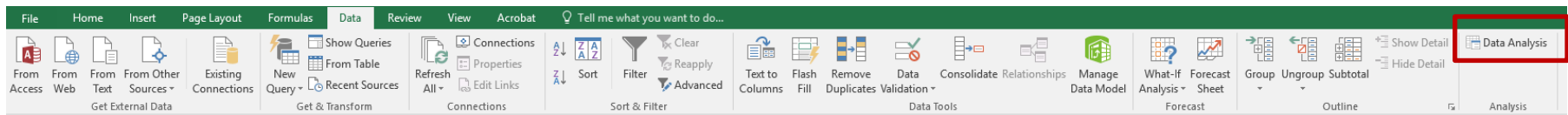
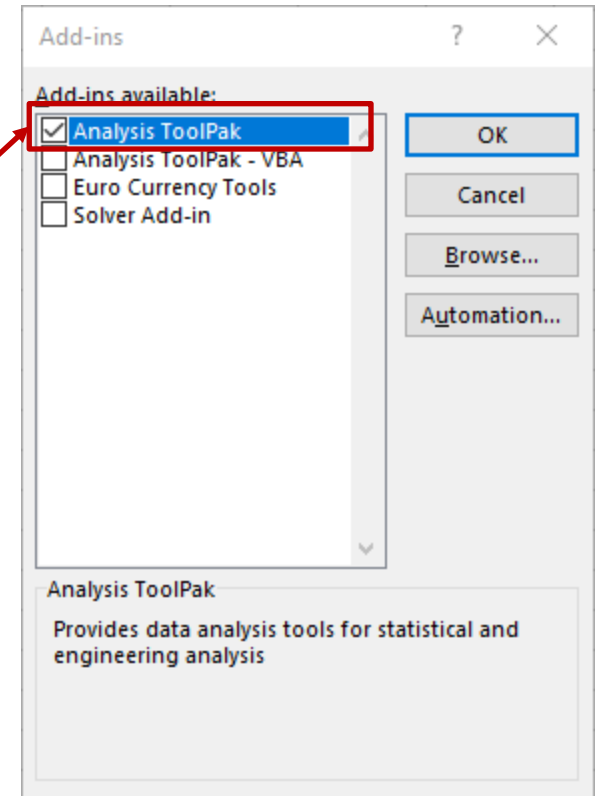
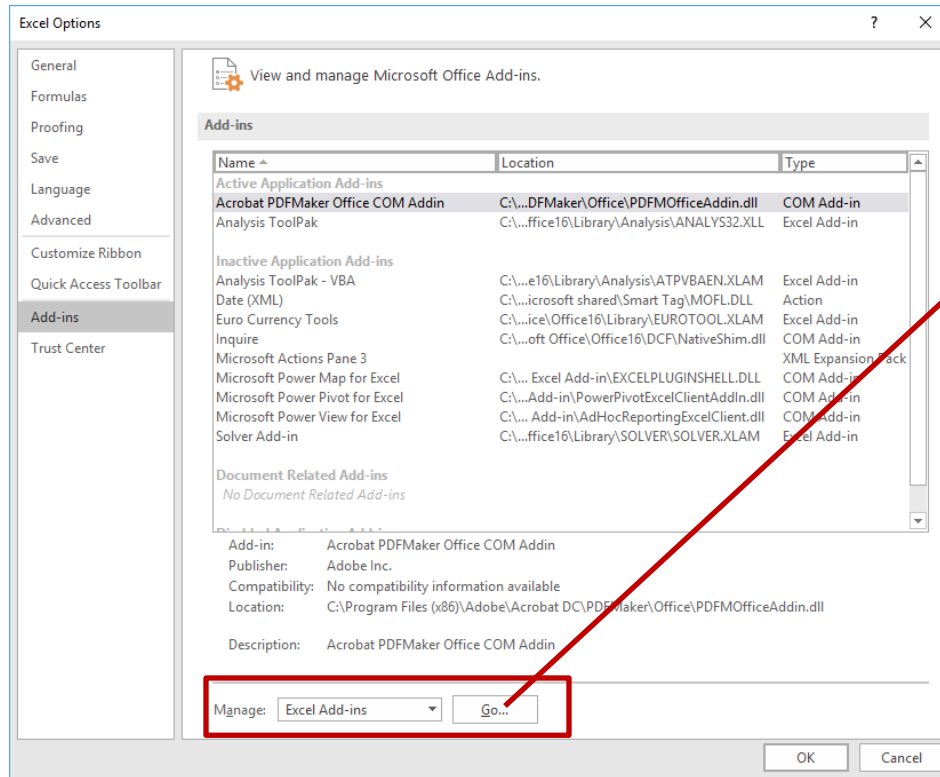
- <https://www.kaggle.com/mohansacharya/graduate-admissions>
- Prediction of Graduate Admissions from an Indian perspective
- 500 instances
- **Manually divided into:**
 - 400 instances training data (Admission_Predict_Ver1.1.train.csv)
 - 100 instances testing data (Admission_Predict_Ver1.1.test.csv)

Software:

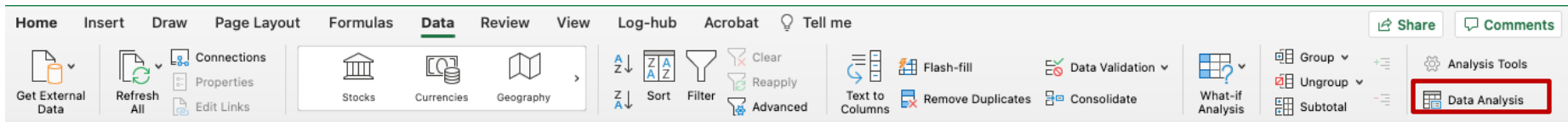
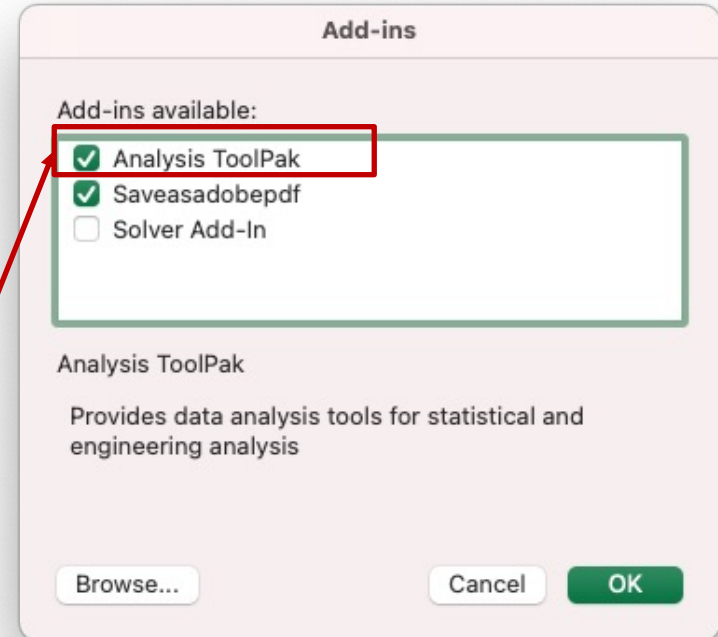
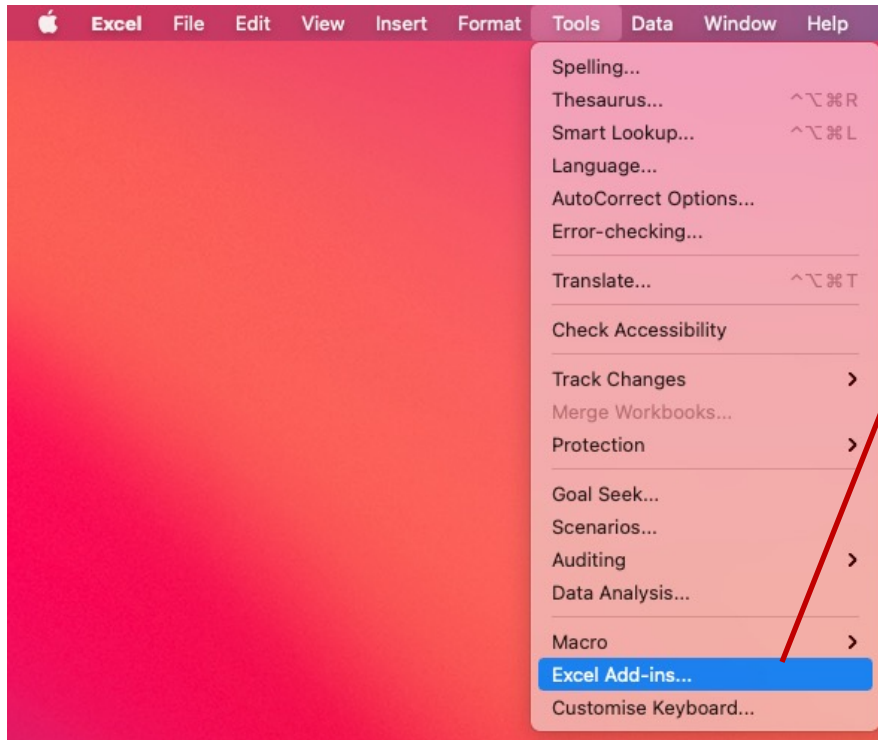
- Excel (with the **Analysis ToolPak**)



INSTALLING EXCEL ANALYSIS TOOLPAK (WIN)



INSTALLING EXCEL ANALYSIS TOOLPAK (MAC)



GENERATING REGRESSION MODEL (EXCEL)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Serial No.	GRE Score	TOEFL Score	University Ra	SOP	LOR	CGPA	Research	Chance of Admit						
2	1	337	118	4	4.5	4.5	9.65	1	0.92						
3	2	324	107	4	4	4.5	8.87	1	0.76						
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5	4	322	110	3	3.5	2.5	8.67	1	0.8						
6	5	314	103	2	2	3	8.21	0	0.65						
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8	7	321	109	3	3	4	8.2	1	0.75						
9	8	308	101	2	3	4	7.9	0	0.68						
10	9	302	102	1	2	1.5	8	0	0.5						
11	10	323	108	3	3.5	3	8.6	0	0.45						
12	11	325	106	3	3.5	4	8.4	1	0.52						
13	12	327	111	4	4	4.5	9	1	0.84						
14	13	328	112	4	4	4.5	9.1	1	0.78						
15	14	307	109	3	4	3	8	1	0.62						
16	15	311	104	3	3.5	2	8.2	1	0.61						
17	16	314	105	3	3.5	2.5	8.3	0	0.54						
18	17	317	107	3	4	3	8.7	0	0.66						
19	18	319	106	3	4	3	8	1	0.65						
20	19	318	110	3	4	3	8.8	0	0.63						

Regression

Input

Input Y Range:

Input X Range:

☒ Labels ☐ Constant is Zero

☐ Confidence Level: %

Output options

☐ Output Range:

☒ New Worksheet Ply:

☐ New Workbook

Residuals

☐ Residuals ☐ Residual Plots

☐ Standardized Residuals ☐ Line Fit Plots

Normal Probability

☐ Normal Probability Plots

OK Cancel

Data Analysis

Analysis Tools

Regression

Sampling

t-Test: Paired Two Sample for Means

t-Test: Two-Sample Assuming Equal Variances

t-Test: Two-Sample Assuming Unequal Variances

z-Test: Two Sample for Means

OK Cancel

Admission_Predict_Ver1.1.train.csv

GENERATING REGRESSION MODEL (EXCEL)

15									
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
17	Intercept	-1.259432478	0.124730747	-10.097209	1.841E-21	-1.5046574	-1.0142076	-1.5046574	-1.0142076
18	GRE Score	0.001737412	0.000597897	2.9058702	0.0038701	0.0005619	0.0029129	0.0005619	0.0029129
19	TOEFL Score	0.002919577	0.001089532	2.6796622	0.0076802	0.0007775	0.0050616	0.0007775	0.0050616
20	University Rating	0.005716658	0.004770425	1.1983539	0.2315032	-0.0036622	0.0150955	-0.0036622	0.0150955
21	SOP	-0.003305169	0.005561643	-0.5942792	0.5526682	-0.0142395	0.0076292	-0.0142395	0.0076292
22	LOR	0.022353127	0.005541485	4.0337793	6.599E-05	0.0114584	0.0332479	0.0114584	0.0332479
23	CGPA	0.118939454	0.012219435	9.7336294	3.382E-20	0.0949156	0.1429633	0.0949156	0.1429633
24	Research	0.024525106	0.007959756	3.0811379	0.0022076	0.008876	0.0401743	0.008876	0.0401743
25									

$-1.259432478 + 0.001737412 * \text{GRE} + 0.002919577 * \text{TOEFL} + 0.005716658 * \text{Uni_Rating} - 0.003305169 * \text{SOP} + 0.022353127 * \text{LOR} + 0.118939454 * \text{CGPA} + 0.024525106 * \text{Research}$

PREDICTING TESTING DATA (EXCEL)

Admission_Predict_Ver1.1.test.csv

$$-1.259432478 + 0.001737412 * \text{GRE} + 0.002919577 * \text{TOEFL} + 0.005716658 * \text{Uni_Rating} - 0.003305169 * \text{SOP} + 0.022353127 * \text{LOR} + 0.118939454 * \text{CGPA} + 0.024525106 * \text{Research}$$

	A	B	C	D	E	F	G	H	I	J
1	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit (Actual)	Chance of Admit (Predicted)
2	401	304	100	2	3.5	3	8.22	0	0.63	0.605305387
3	402	315	105	2	3	3	8.34	0	0.66	0.654940123
4	403	324	109	3	3.5	3	8.94	1	0.78	0.782207991
5	404	330	116	4	4	3.5	9.23	1	0.91	0.862802581
6	405	311	101	3	2	2.5	7.64	1	0.62	0.575424919
7	406	302	99	3	2.5	3	7.45	0	0.52	0.516349434

Predicted values

RECALL: TYPES OF DECISION PROBLEM (LEARNING PROBLEMS)

Regression (Supervised)

- Stock price prediction

Classification (Supervised)

- Weather forecast (sunny, rainy, cloudy, etc)

Clustering (Unsupervised)

- Group Facebook users based on their interest

CLASSIFICATION

Classification (Supervised)

- Using existing data instances (labeled) to learn a model for predicting subsequent instances (unlabeled)
- Example:
 - Assume that you have some attributes of the current weather, and we need to decide whether to go out to play

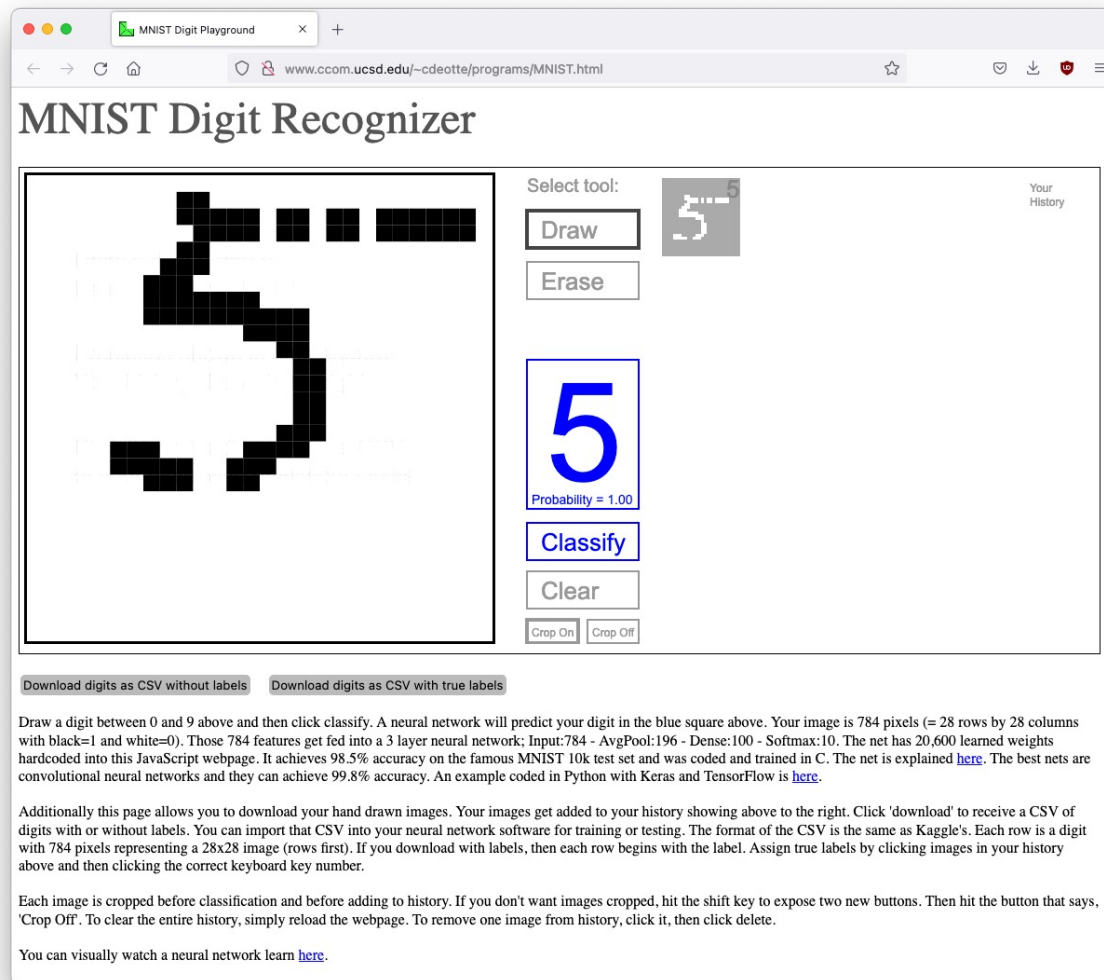
No.	1: outlook Nominal	2: temperature Nominal	3: humidity Nominal	4: windy Nominal	5: play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

Training
using an
algorithm

Classifier

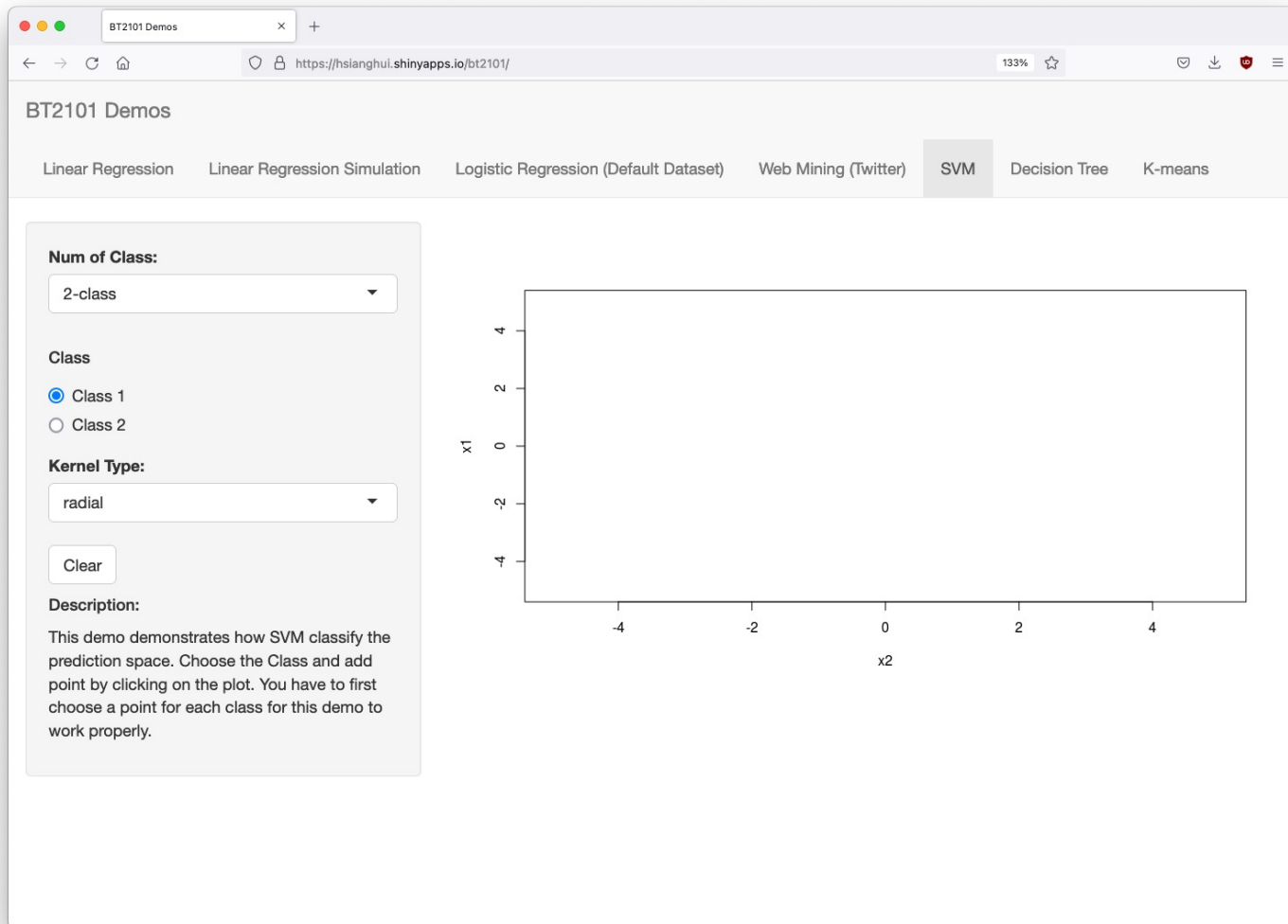
outlook: rainy
temperature: mild
humidity: high
windy: TRUE
play: ?

CLASSIFICATION DEMO



<http://www.ccom.ucsd.edu/~cdeotte/programs/MNIST.html>

CLASSIFICATION DEMO



<https://hsianghui.shinyapps.io/bt2101/>

UNSUPERVISED LEARNING

Introduction to
Web Mining

Why choose
this course?

Supervised
Learning

Unsupervised
Learning

Web Scraping
Demo

MACHINE LEARNING

I don't really care how you do it
but when you see these
values, this is how you should do
prediction.



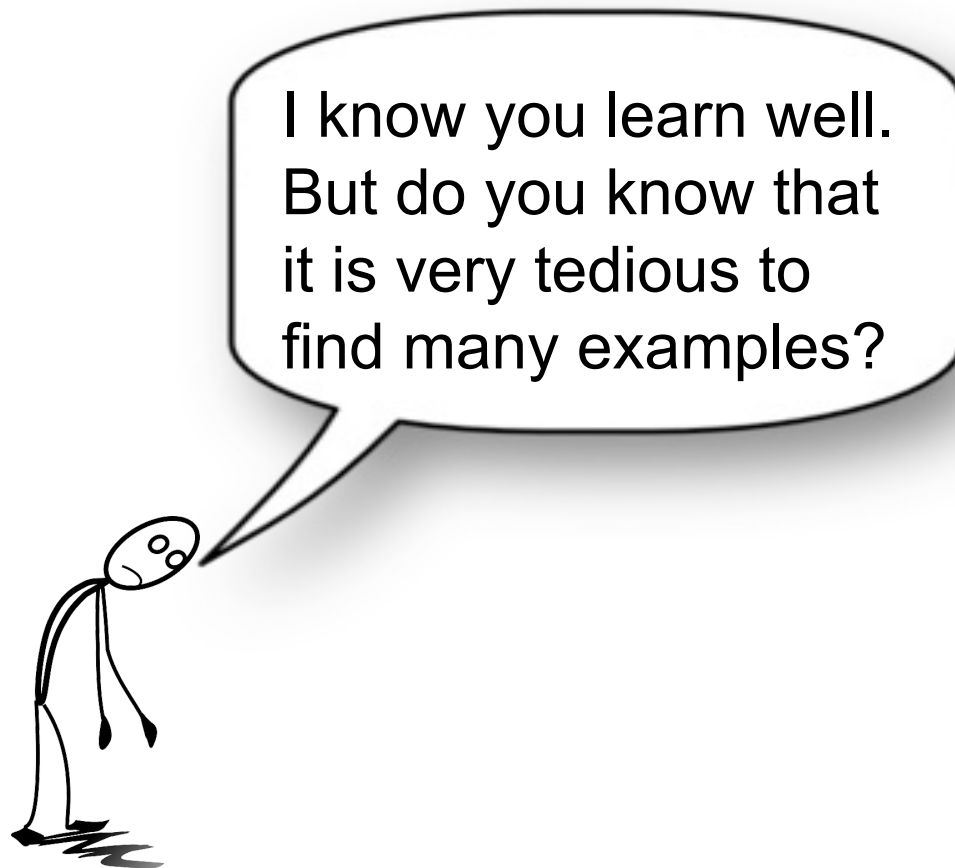
MACHINE LEARNING

Ok I've learnt something about the data but I get better at this if you give me more examples!



Machine

CHALLENGES OF SUPERVISED LEARNING



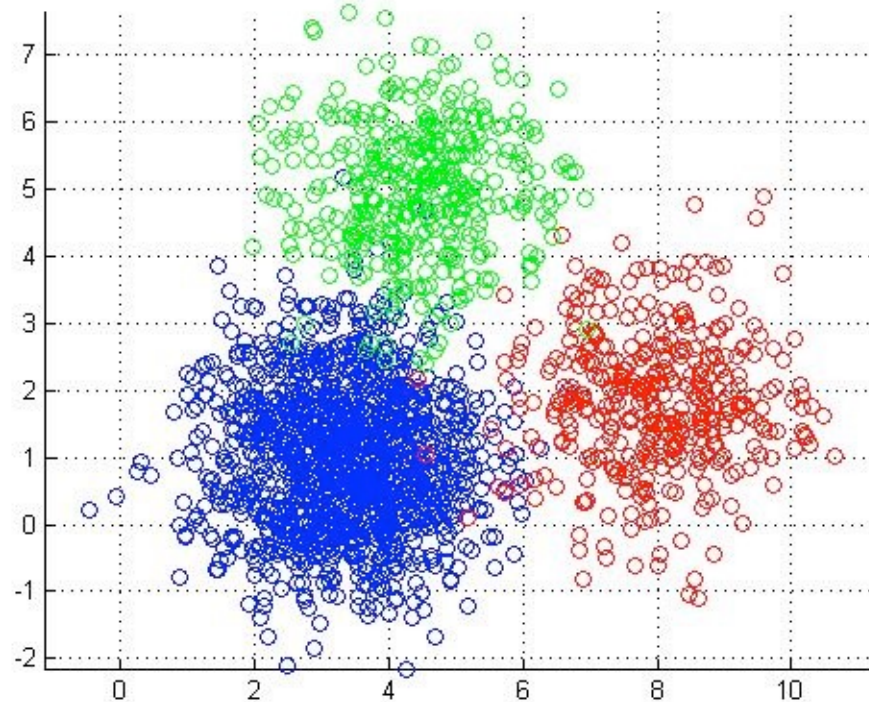
UNSUPERVISED LEARNING

Unsupervised

- Assume we only have unlabeled data and we want to either:
 - label the data instances
 - or group up data instances into subsets sharing common characteristics
- One commonly application of unsupervised approach is clustering
- What do you think is the accuracy of unsupervised approaches compared to supervised?
- Why are unsupervised approaches important?

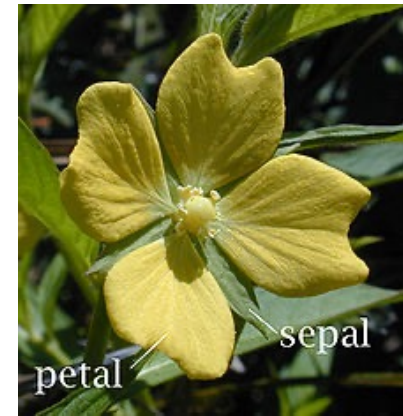
CLUSTERING EXAMPLE

Points are clustered together because they are closer to each other



UNSUPERVISED

Clustering Example (Iris flower)



Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
6.2	2.2	4.5	1.5	versicolor
5.7	2.6	3.5	1	versicolor
5.5	2.4	3.8	1.1	versicolor
5.5	2.4	3.7	1	versicolor
7.2	3.6	6.1	2.5	virginica
6.5	3.2	5.1	2	virginica
6.4	2.7	5.3	1.9	virginica
6.8	3	5.5	2.1	virginica



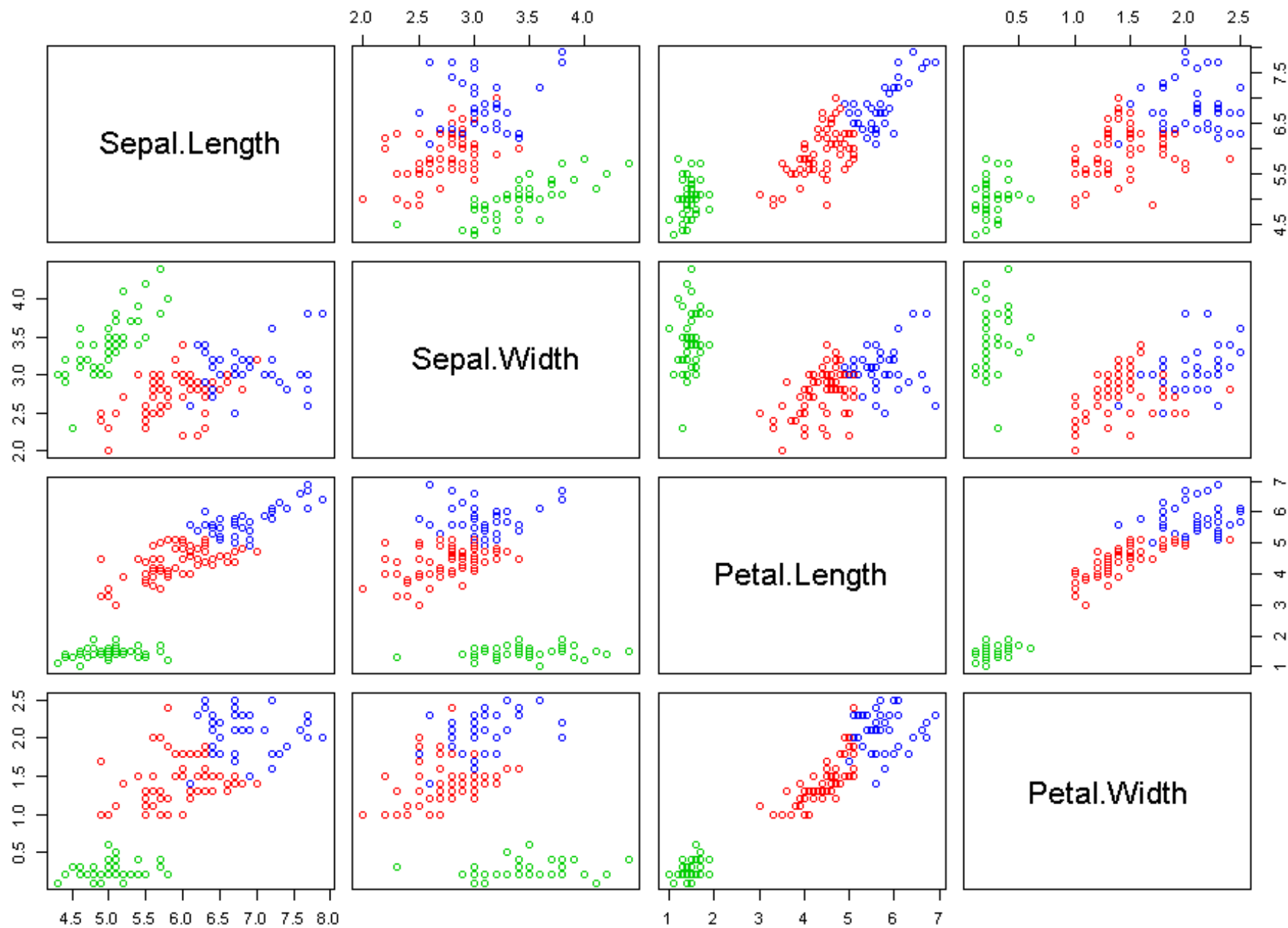
Iris Setosa



Iris Versicolor



Iris Virginica



OTHER TYPES OF DECISION PROBLEMS

Regression (Supervised)

- Stock price prediction

Classification (Supervised)

- Weather forecast (sunny, rainy, cloudy, etc)

Clustering (Unsupervised)

- Group Facebook users based on their interest

OTHER TYPES OF DECISION PROBLEMS

Data mining techniques:

- Regression
- Classification
- Clustering
- Association Rules
- Neural Networks
- Deep Learning
- etc



Already discussed briefly
earlier

ASSOCIATION RULES

Association Rules

- Quite different from Regression and Classification
- Still using data but there is no clear-cut prediction that we can derive
- Also different from Clustering in that we are clear in our objectives
- We are interested in mining **Association Rules** which look like:
 - $X \rightarrow Y$ (X implies Y)
 - E.g. {diaper, milk} $\rightarrow$ {beer}

SUPERMARKET PROBLEM

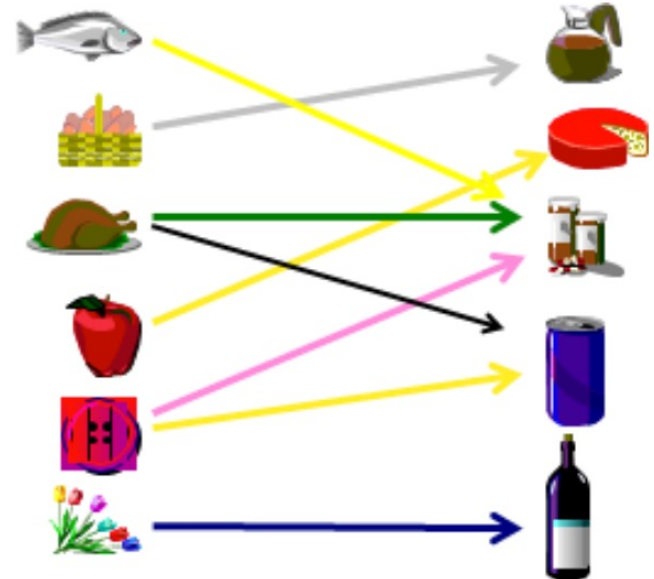


One of the key problems that a supermarket wants to tackle is how to place the products

Place similar category products together for better searching?

MARKET BASKET ANALYSIS

Market Basket Analysis (MBA) aims to find **association** between groups of items based on the **transactions**



98% of people who purchased items A and B also purchased C

SUPERMARKET PROBLEM



Probably better to place products that users are likely to buy together in the same location

- Likely to generate more sales that way

WEB SCRAPING DEMO

Introduction to
Web Mining

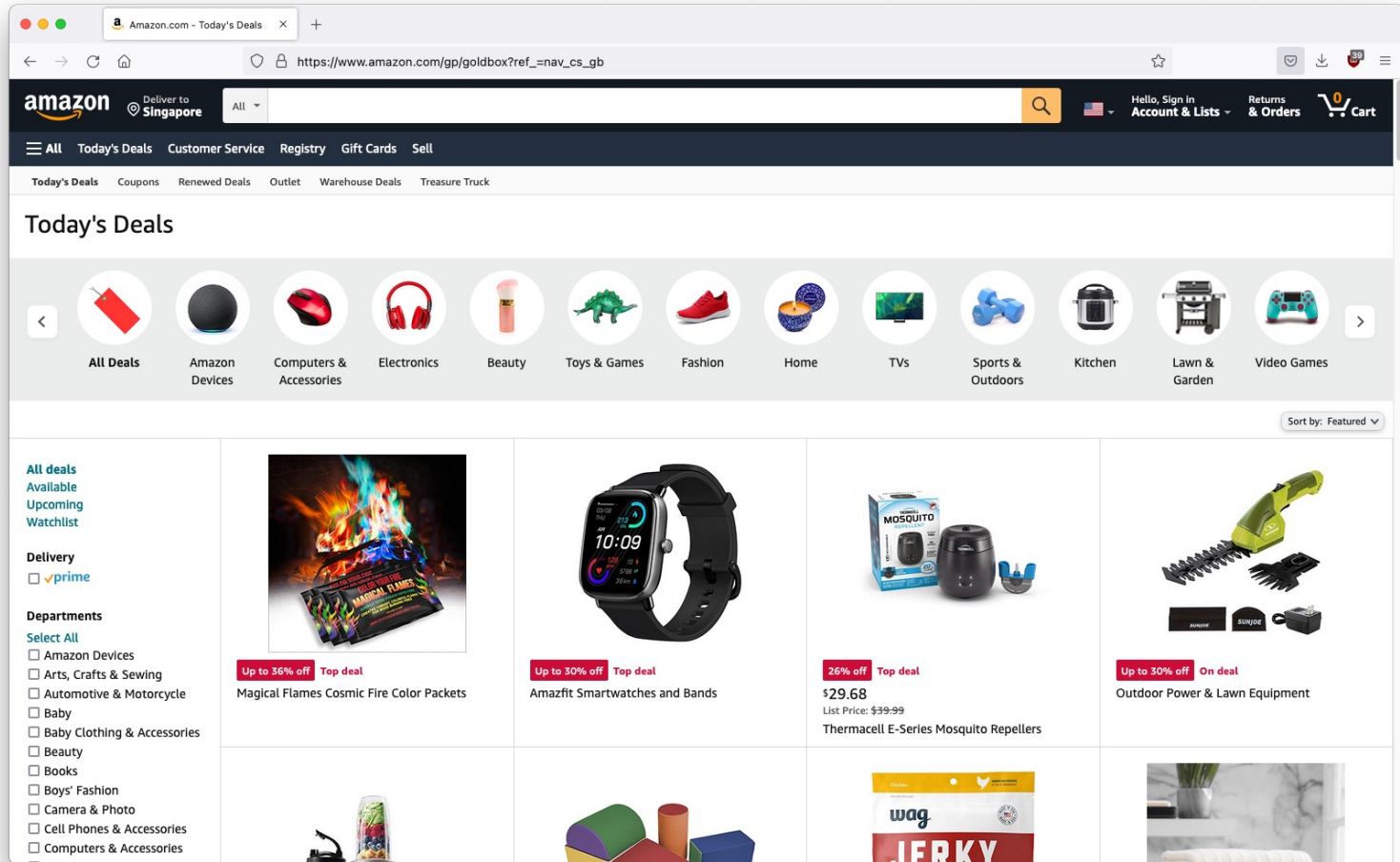
Why choose
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Demo

WEB SCRAPING DEMO: AMAZON.COM



CHALLENGE

Are you able to get the review text of a product?

Amazon.com: Customer reviews: X +

← → ↺ 🏠 🔒 https://www.amazon.com/Apple-iPhone-XR-64GB-White/product-reviews/B08BGD4G36 120% ☆ 📧 ⬇️ 🛡️ ☰

Top reviews ▾ All reviewers ▾ All stars ▾ All formats ▾ Text, image, video ▾

45,886 total ratings, 9,390 with reviews

From the United States

Alan marnik

★★★★★ **As advertised**

Reviewed in the United States on April 27, 2019

Size: 128GB | Color: Blue | Service Provider: Unlocked | Product grade: Renewed | **Verified Purchase**

I was a little sceptical buying this phone because some of the reviews said that the phone they received was locked or had slight damage. However, the one I received was perfect. It came fully unlocked and undamaged. It looks as if I had bought it straight from apple. So far an overall 5 stars and no complaints.

1,125 people found this helpful

Helpful Report abuse

Diamond Lovemore

★★★★★ **Trustworthy**

Reviewed in the United States on June 12, 2019

Size: 64GB | Color: Coral | Service Provider: Unlocked | Product grade: Renewed | **Verified Purchase**

This seller is amazing was a bit skeptical at first but this phone met my every expectation, very trustworthy in every way no it doesn't come with headphones. At least it comes with a charger & a 90 day warranty I also got a brand new spanking battery I've checked the serial number online it is worthy!!



942 people found this helpful

Helpful Report abuse

griffin cannon

★★★★★ **FRAUDS- phones SOLD ARE STOLEN**

Reviewed in the United States on August 20, 2019

Size: 64GB | Color: Black | Service Provider: AT&T | Product grade: Renewed | **Verified Purchase**

Just was told by my carrier, ATT, that this phone is blacklisted. The phone was blacklisted after 1 week of receiving the phone. THANKS AMAZON

698 people found this helpful

Helpful Report abuse

CHALLENGE

And use a word cloud generator to summarize the insights

The screenshot shows the 'Free Word Cloud Generator' website. The 'Paste/Type Text' section contains a text area with the following text: 'tablet as gift for my birthday. Right off the top it wouldn't read the SD memory.. The software was setup to load everything personal into areas that are best kept private. Admittedly it usually takes time to learn and get use to new software but there was the problem with the SD memory. I returned it for an Apple 9th generation 10.2 tablet.. I couldn't believe how easy it was to get started.. also I didn't have to find and download an app that would transfer data from my iPhone.. Just easier that's all. I'm very pleased with this device and would tell anyone to go for it.'

The 'Options' section includes a 'Word List' dropdown, a 'Font' dropdown, a 'Colors' dropdown, and a 'Download' button. A progress bar is set to 100. There are 'CLEAR' and 'VISUALIZE' buttons.

The generated word cloud features the following words: android, know, although, longest, among, birthday, overall, whiteboard, least, brand, always, believe, experience, works, iphone, useful, ipad, freeform, great, purchase, thought, time, sd, price, highly, little, getting, work, top, apple, generation, years, supported, wi-fi, pricing, new, good, ios, otterbox, gbcolor, defender, case, old, air, base, style, looking, condition, ipads, mine, silver, verified, use, design, wifisize, purchased, device, reason, longer, pencil, actually, first, working, one, memory, tablet, need, go, performance, app, pen, gb, recommend, cpu, refund, excellent, better, sold, say, definitely, capacity, space, review, gray, verified, bought, purchase, return, model, basic, sale, replacement, features, wonderfully, data, brainstorming, anyone, box, laptop, input.

<https://www.freewordcloudgenerator.com/generatwordcloud>

WHAT'S NEXT?

Cross-Industry Standard Process for Data Mining (CRISP-DM) & Predictive Analytics I